MEDS Module 4 Day 3

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TASK 3: Half Adder and Full Adder: assign vs always_comb

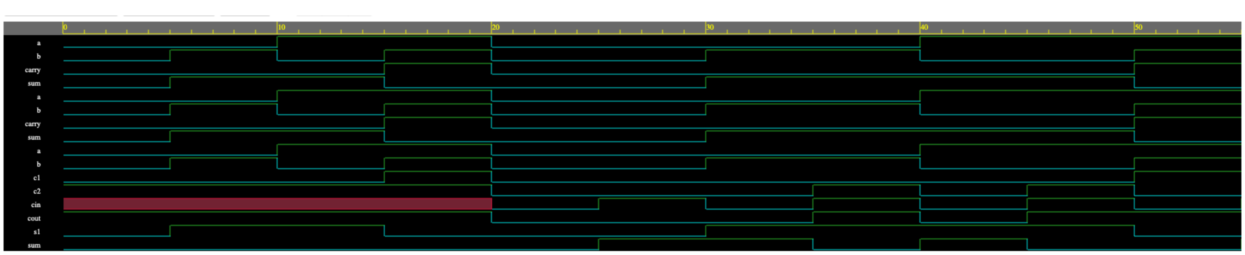
1. EDA Playground Link: https://edaplayground.com/x/gHmi

2. Terminal Output Screenshot:

```
Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64;  Jul 22 17:19 2026
Half Adder Comparison
---------------------------------------------
a b | assign(sum carry) | always(sum carry)
0 0 |      0    0      |      0    0
0 1 |      1    0      |      1    0
1 0 |      1    0      |      1    0
1 1 |      0    1      |      0    1

Full Adder Truth Table
--------------------------------
a b cin | sum cout
0 0  0  |  0    0
0 0  1  |  1    0
0 1  0  |  1    0
0 1  1  |  0    1
1 0  0  |  1    0
1 0  1  |  0    1
1 1  0  |  0    1
1 1  1  |  1    1
$finish called from file "design.sv", line 73.
$finish at simulation time                    60
```

3. Waveform Screenshot:



Note: To revert to EPWave opening in a new browser window, set that option on your profile page.

4. Explanation: Both half adders are synthesized to the same logic gates because the functionality of both is the same.

TASK 5: Parity Bit Generator

1. EDA Playground Link: https://edaplayground.com/x/ut8c

2. Terminal Output Screenshot:

```
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64;  Jul 22 17:32 2026
Data  Parity Expected Result
----------------------------
0000     0        0       PASS
0001     1        1       PASS
0010     1        1       PASS
0011     0        0       PASS
0100     1        1       PASS
0101     0        0       PASS
0110     0        0       PASS
0111     1        1       PASS
1000     1        1       PASS
1001     0        0       PASS
1010     0        0       PASS
1011     1        1       PASS
1100     0        0       PASS
1101     1        1       PASS
1110     1        1       PASS
1111     0        0       PASS
$finish called from file "design.sv", line 33.
$finish at simulation time                   80
```

3. Waveform Screenshot: